



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 146



BAYER PRESSURE TANK

BAYER PRESSURE TANK FOR FLAME THROWER

Test Report No.	:	2013 - 146
Date of Test	:	October 4 & 7, 2013
Test Valid Until	:	October 7, 2013
Brand	:	No data
Model	:	No data
Type	:	Cylindrical, compression tank
Manufacturer	:	No data
Test Requested by	:	BAYER CROP SCIENCE PHILS., INC. Masiit, Calauan, Laguna
Machine Classification	:	Commercial

SUMMARY

The pressure tank used for flame thrower was tested for leakage and durability.

Leakage test. The pressure tank was initially pressurized to 90, 100, and 120 psi pressure, left for eight hours and the final pressure was recorded. For the initial pressure of 90 psi, the reduction in pressure after eight hours was five psi (5.56%). For the 100 psi initial pressure, the pressure reduction after eight hours was eight psi (8.0%). At initial pressure of 120 psi, the reduction in pressure was 10 psi (8.33%) after eight hours.

Durability Test. The tank was mounted on a test rig and was subjected to stress and stress-free condition. It was subjected to alternate 0 and 80 psi. No breakdown occurred after 200 pressure cycles.

PURPOSE AND SCOPE OF TEST

The pressure tank was tested for leakage and strength. The test was conducted at AMTEC on October 4 & 7, 2013. The test was carried out to verify and assess the following:

1. Tank leakage
2. Tank strength

METHOD OF TEST

The sprayer was tested based on the Philippine Agricultural Engineering Standard “PAES 113:2000 Agricultural Machinery – Lever-Operated Knapsack Sprayer - Methods of Test”.

OPERATOR’S MANUAL

No operators manual was submitted.

DESCRIPTION OF THE MACHINE

The pressure tank (**Figure 1**) was a cylindrical steel tank provided with valve stem (**Figure 2**) as an air inlet opening and an outlet opening (**Figure 3**) for pressure hose connected to the flame outlet. It was provided with filler hole (**Figure 4**) for liquid filling. It was used as pressure tank for the flame thrower. An ordinary manual pressure pump capable of producing pressure enough for the purpose could be used to pressurize the tank through the valve stem. The valve stem functions also as a check valve. It had a pressure gage to monitor the pressure inside the tank. The tank was also provided with shoulder strap.

Detailed specifications of the machine are found in **Table 1**.



Figure 1. BAYER Pressure Tank



Figure 2. Valve stem

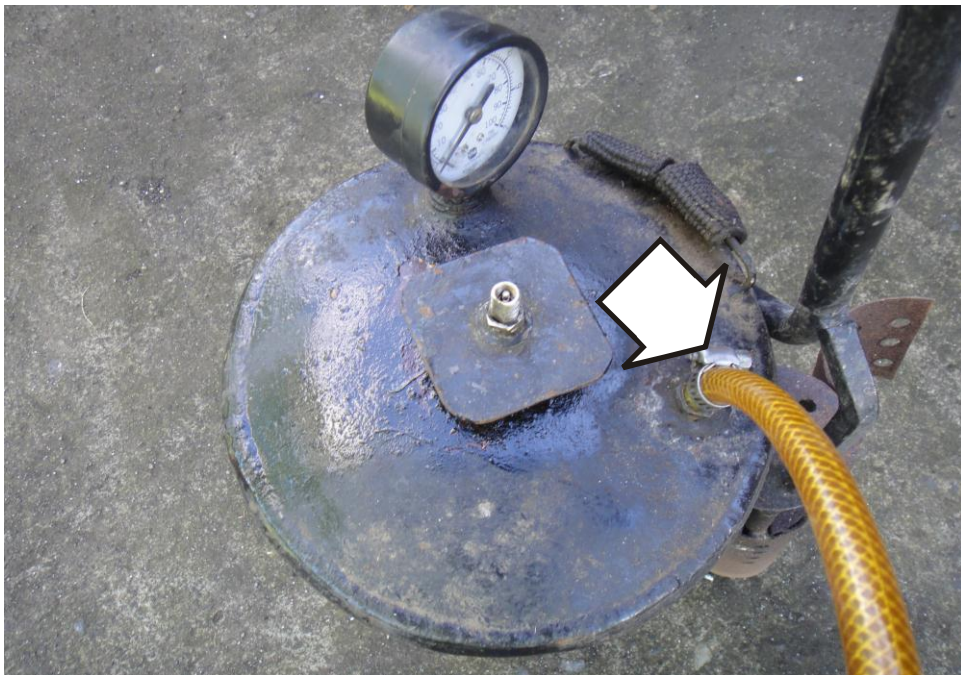


Figure 3. Outlet opening



Figure 4. Tank's filler hole

SPECIFICATIONS

Table 1. Detailed specifications of the BAYER pressure tank

A.	Brand	:	No data
B.	Model	:	No data
C.	Type	:	Cylindrical, Compression
D.	Make	:	No data
E.	Dimensions (tank only) and Weight		
1.	Height (mm)	:	500
2,	Outside Diameter (mm)	:	305
3.	Weight (kg)	:	7.10
F.	Normal working pressure (kg/cm ²)	:	No Data
G.	Diameter of filler hole (mm)	:	26
H.	Material	:	Steel

RESULTS AND DISCUSSIONS

A. LEAKAGE TEST

The test was carried out in the laboratory to determine the effectiveness of the different seals and the presence of any pin hole on the tank. The tank was pressurized to 90, 100, and 120 psi pressures and left for eight hours. Results showed that at 90 psi initial pressure the pressure left after eight hours was 85 psi, 92 psi for the 100 psi, and 110 psi for the 120 psi pressures. This was 7.30% average pressure reduction for eight hours.

Please refer to **Table 2**.

B. DURABILITY TEST

The tank was mounted on the test set-up as shown in **Figure 5**. It was tested to determine the strength of the tank. It was subjected to an alternate stress and stress-free condition for 200 pressure cycles. The pressure applied was 80 psi and reduced to zero as one pressure cycle, this was done up to 200 pressure cycles. No breakdown occurred during the test.

Please refer to **Table 3**.



Figure 5. The tank on the durability test set-up

PERFORMANCE TEST

Machine Tested : BAYER Pressure Tank
Location of Test: AMTEC Test Laboratory
U.P. Los Baños
Date of Test : October 4 & 7, 2013

Table 2. Results of Leakage Test

ITEMS	Data after Eight hours		
	Final pressure (psi)	Pressure reduction (psi)	Percent pressure reduction
90	85	5	5.56
100	92	8	8.0
120	110	10	8.33

Table 3. Results of Durability Test

Pressure cycles	Initial pressure (psi)	Maximum pressure (psi)	Remarks
120	0	80	No breakdown occurred

Test Engineer:

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DELFIN C. SUMINISTRADO
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Date

N.B.

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